Data Technician Week 2 Workbook Tableau and Power BI

Name: Hamidah Ali

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Day 1: Task 1

Please research the different versions of Tableau, compare them below and explain the limited functionality on 'Tableau Public'.

Different Tableau versions

Tableau version:	Main purpose:	Key features:	Limitations:
Tableau public	Learning, personal projects and publicly sharing visualisations.	Free; create dashboards and visualisations; publish and share online; build a public portfolio.	Data and workbooks published are public; limited data connections; maximum 15 million rows per workbook; limited data refresh; not intended for commercial/private use.
Tableau desktop free edition	Personal or business data analysis without online sharing.	Full Tableau Desktop analytics; can work with private data; live connections; no 15-million-row limit.	Cannot publish/share work online or connect to Tableau cloud/ server.
Tableau desktop professional	Professional/business data analytics and visualisation.	Connects to a wide range of data sources; live data; publish to Tableau cloud or server; supports sharing and collaboration.	Requires paid licence.
Tableau cloud	Cloud-based analytics and sharing for organisations	Fully hosted; create, publish, manage and share dashboards/data online; no organisation-managed server required.	Paid product and designed primarily for organisational use.
Tableau server	Hosting and sharing Tableau content within an organisation	Self-hosted; organisations can control their own environment, users, data and permissions.	Requires organisation-managed infrastructure and administration.

Day 1: Task 2 (Optional)

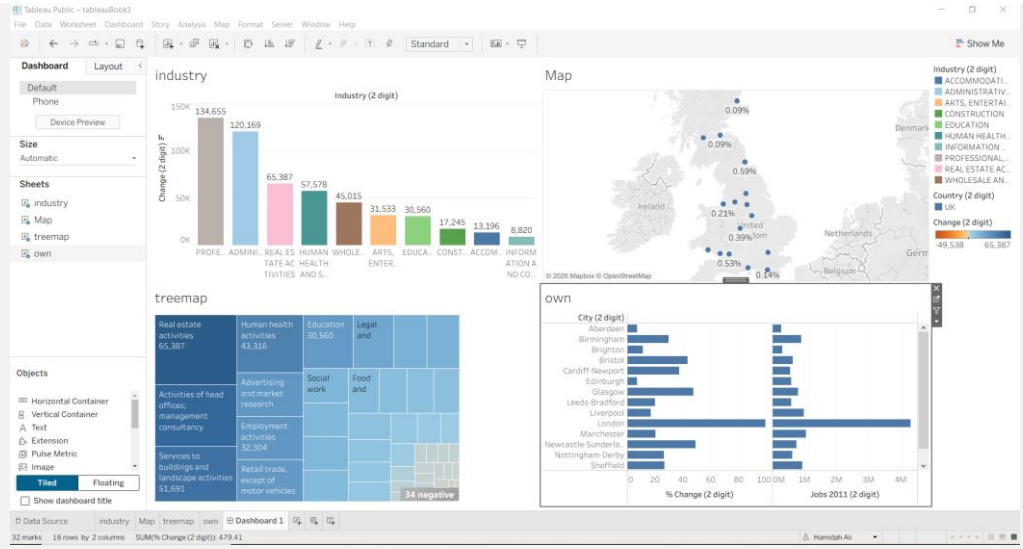
Please access and complete the DataCamp course.

Introduction to Tableau - DataCamp Learn

Paste your XP here.	XP2400

Day 1: Task 3

Please attach the screenshot and link to your **EMSI Job Change** (GB Employment) dashboard below.

Paste your print screen here	
Link to your dashboard	https://public.tableau.com/views/tableauBook3_17900673093170/Dashboard1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Day 2: Task 1

Start new Tableau page.
Download GapminderHealth.xlsx dataset.
Using the Health GapminderHealth dataset, conduct an analysis to find trends and key information that could be used by an organisation for future support.

Column Name	Description	Data Type	Example

Continent	Continent where the country is located.	Text	Asia
Country	Name of the country.	Geographic	Afghanistan
Year	Year of observation or measurement.	Numeric	1990
Life Expectancy	Average number of years a person is expected to live at birth.	Numeric	48.9
Gender	Gender group for which the metrics are reported.	Text	Women
BMI	Average Body Mass Index of the population.	Numeric	20.7
Blood Pressure	Average blood pressure measurement of the population.	Numeric	124
Cholesterol	Average cholesterol level of the population.	Numeric	4.55
Lung Cancer	Lung cancer incidence	Numeric	194
Liver Cancer	Liver cancer incidence	Numeric	107
Stomach Cancer	Stomach cancer incidence	Numeric	406
Population	Total population of the country for the specified year.	Numeric	6,067,694
Population Growth	Annual population growth rate expressed as a percentage.	Decimal (%)	6.75

Scenario:
You are a data analyst working for a global health organization. Your team needs to quickly understand key health trends and disparities across different countries and continents. They are particularly interested in seeing how health metrics vary and how life expectancy has changed over time.

Your Task:
Using the GapminderHealth dataset in Tableau, create a series of visualizations to answer the following questions. Focus on using basic chart types and features.

Instructions & Deliverables:

1. **Life Expectancy by Continent (Most Recent Year):**
    - Create a **bar chart** showing the average Life Expectancy for each Continent.
    - **Filter** the data to show only the **most recent year** available in the dataset.
    - **Sort** the continents by Life Expectancy in ascending or descending order.
    - **Deliverable:** A worksheet named "Life Expectancy by Continent".

2. **Life Expectancy Trend Over Time (Top 5 Countries):**
    - Create a **line chart** to show the trend of Average Life Expectancy over Year.
    - **Filter** the data to display only the **top 5 countries** with the highest average Life Expectancy across all years. (Hint: Use a Top N filter on Country based on Average Life Expectancy).
    - Use Country on Colour to differentiate the lines.
    - **Deliverable:** A worksheet named "Life Expectancy Trend".

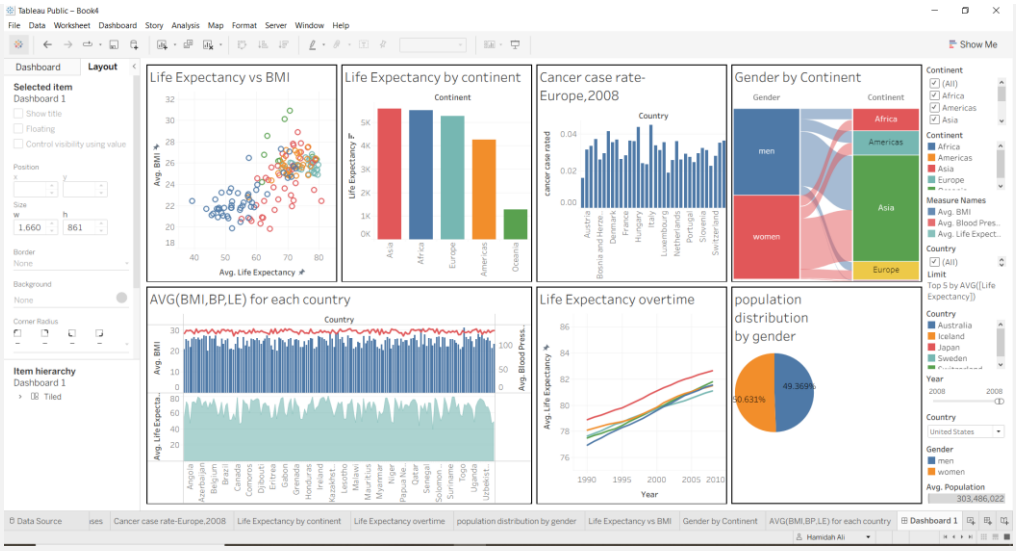
3. **Population Distribution by Gender (Latest Year for a Selected Country):**
    - Create a **pie chart** showing the Population distribution by Gender.
    - **Add a Country filter** to the worksheet (e.g., set it to 'United States' or a country of your choice initially, but ensure the filter is available for interaction). Change filter type to Single Value (dropdown).
    - **Filter** the data to show only the **most recent year** available in the dataset.

- **Add Population to Labels.** Right click on sum (Population) under marks card and change **Quick table calculation to Percentage of Total.**
- **Deliverable:** A worksheet named "Population by Gender".

4. **Health Metric Comparison (Scatter Plot):**
   - Create a **scatter plot** to explore the relationship between Average Life Expectancy and Average BMI.
   - Place Life Expectancy on one axis and BMI on the other.
   - Use Country on Detail to represent individual countries as points.
   - Use Continent on Colour to differentiate points by continent.
   - Add Continent to filters and show filter.
   - **Deliverable:** A worksheet named "Life Expectancy vs BMI".
5. **Viz of your Choice**
   - Use any other fields or chart types to add another visual.

6. **Dashboard Creation:**
   - Combine all worksheets (Life Expectancy by Continent, Life Expectancy Trend, Population by Gender, Life Expectancy vs BMI, Viz of your Choice) into a **single dashboard**.
   - Ensure the "Country" filter on the "Population by Gender" worksheet applies to the "Life Expectancy Trend" worksheet as well (if applicable or make it a global filter if you prefer the user to select one country to see all relevant details for it). *Self-correction: Given it's basic, let's keep the filters simple. Just place the filter on the dashboard and ensure it's applied to the relevant sheet(s) where it makes sense.*
   - Add a **Title** to your dashboard, e.g., "Global Health Insights".
   - Publish your dashboard to Tableau Public Website

- **Deliverable:** A dashboard named "Global Health Dashboard".

Paste your dashboard screens here	
What did you find and any reflections on how the NHS could use this?	**The dashboard highlighted differences in health measures across countries, continents, genders and over time. I found that life expectancy generally increased over the period shown, although the trends varied between continents. The BMI and life expectancy scatter plot also showed variation between countries, while the cancer case rate chart demonstrated differences between countries in Europe. The population distribution chart showed a relatively similar proportion of men and women in the selected population.** **This type of dashboard could be useful to the NHS for monitoring population health and identifying trends or differences between demographic groups and geographical areas. Interactive dashboards could help healthcare organisations identify areas where particular health issues may require further investigation, support public health planning and help allocate resources based on population needs. For example, trends in BMI, life expectancy or cancer rates could help inform preventative health programmes and highlight areas for further research.** **However, I would not use this dataset on its own to make clinical or policy decisions. The data would need to be current, accurate and representative of the relevant NHS population. Additional information, such as socioeconomic factors, ethnicity, age, healthcare access and clinical outcomes, would also be important when interpreting health inequalities.**
Link to your dashboard	https://public.tableau.com/views/globalhealthinsights_17900831976430/Dashboard1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Once you finish this task save and close Tableau.

Day 2: Task 2

Start Tableau Public Desktop and import the Spotify dataset SpotifyFeatures.xlsx, conduct an analysis to find trends and key information that could be used by an organisation for future projects.

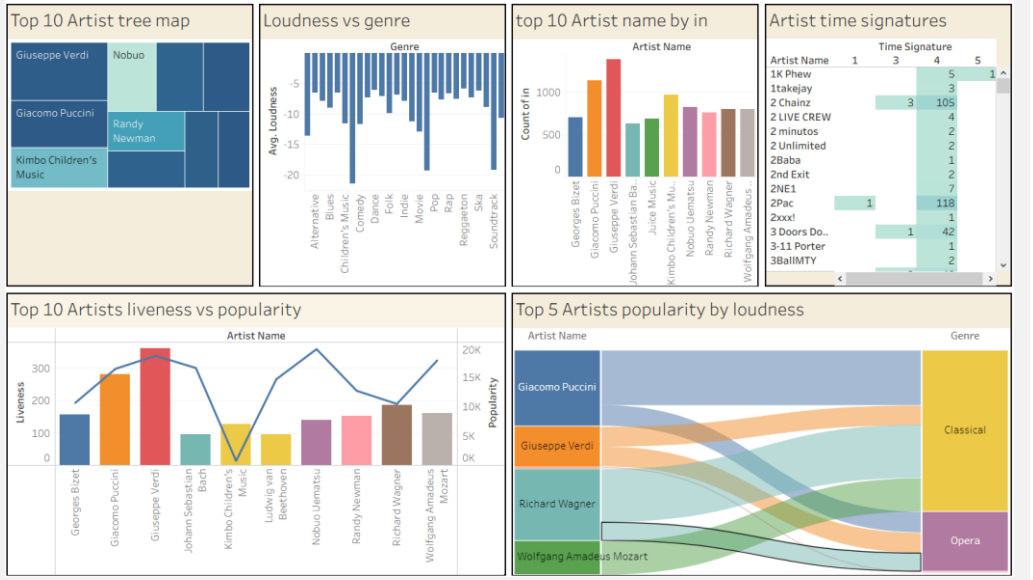
Column Description

- **genre**: The genre in which the track belongs
- **artist_name**: The artists' names who performed the track. If there is more than one artist, they are separated by a;
- **track_name**: Name of the track
- **track_id**: The Spotify ID for the track
- **popularity**: **The popularity of a track is a value between 0 and 100, with 100 being the most popular**. The popularity is calculated by algorithm and is based, in the most part, on the total number of plays the track has had and how recent those plays are. Songs that are being played a lot now will have a higher popularity than songs that were played a lot in the past. Duplicate tracks (e.g. the same track from a single and an album) are rated independently. Artist and album popularity is derived mathematically from track popularity.
- **acousticness**: A confidence measure from 0.0 to 1.0 of whether the track is acoustic. 1.0 represents high confidence the track is acoustic
- **danceability**: Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable
- **duration_ms**: The track length in milliseconds
- **energy**: Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale
- **instrumentalness**: Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal". The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content
- **key**: The key the track is in. Integers map to pitches using standard Pitch Class notation. E.g. 0 = C, 1 = C♯/D♭, 2 = D, and so on. If no key was detected, the value is -1

- **liveness**: Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live
- **loudness**: The overall loudness of a track in decibels (dB)
- **mode**: Mode indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0
- **speechiness**: Speechiness detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks
- **tempo**: The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration
- **time_signature**: An estimated time signature. The time signature (meter) is a notational convention to specify how many beats are in each bar (or measure). The time signature ranges from 3 to 7 indicating time signatures of 3/4, to 7/4.
- **valence**: A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry)

There is no set scope for the analysis, simply to find trends and document them below:

Paste your print screens here



What did you find?	I found that classical music artists feature strongly in the dataset, with artists such as Giuseppe Verdi, Giacomo Puccini and Georges Bizet appearing among the most represented artists. The loudness chart shows that most genres have relatively low average loudness, although there are differences between genres. I also found that popularity and liveliness vary between artists; for example, Giuseppe Verdi has one of the highest liveliness values and also has high popularity. The final chart shows that the most popular artists are mainly associated with Classical and Opera genres. Overall, the dashboard shows differences in artist popularity, liveliness and loudness across different genres.
Link to your dashboard	https://public.tableau.com/views/Musicinformation/Dashboard1?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

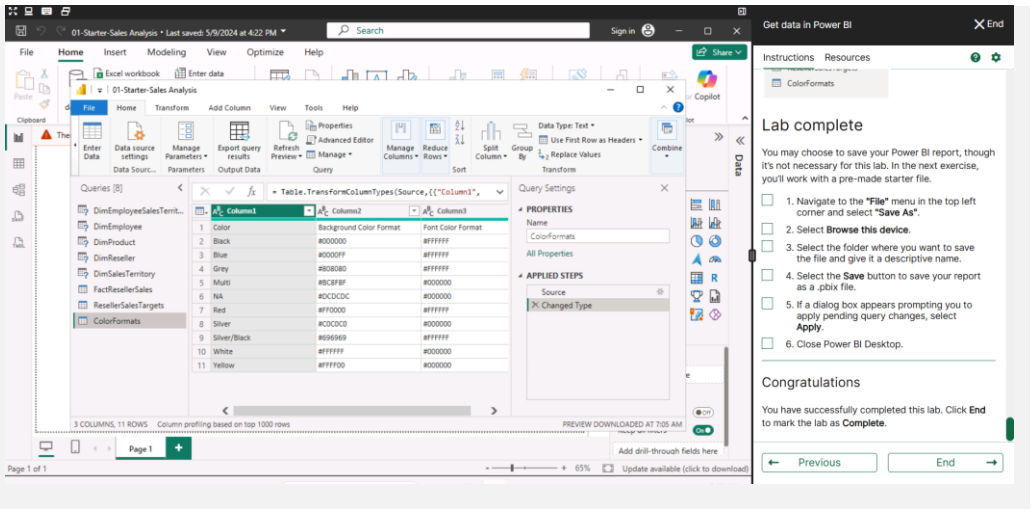
Day 3: Task 1

Please complete **Get data in Power BI** on Skillable. Once complete, paste a print screen below and in the collaboration board.

1) Visit https://leeptalent.learnondemand.net/
2) Sing into your profile

3) Go to **PL-300T00-A Design and Manage Analytics Solutions Using Power BI**
4) Launch and complete **Lab 1: Get data in Power BI**

Paste your completed lab here

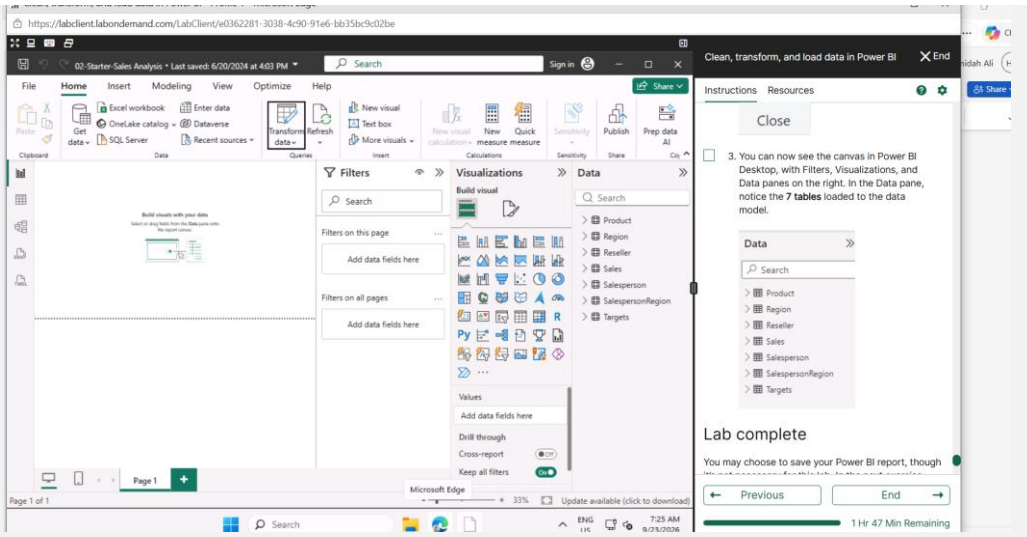


Day 3: Task 2

Please complete **PL-300T00-A-Lab-02-LA] Clean, transform, and load data in Power BI** on Skillable. Once complete, paste a print screen below and in the collaboration board.

1) Visit https://leeptalent.learnondemand.net/
2) Sing into your profile
3) Go to **PL-300T00-A Design and Manage Analytics Solutions Using Power BI**
4) Launch and complete **Lab 2: Clean, transform, and load data in Power BI**

Paste your completed lab here



Day 3: Task 3 (Optional)

Explore Power BI course on DataCamp if you finish Day 3 Task 2 early.

Introduction to Power BI

Paste your completed XP here

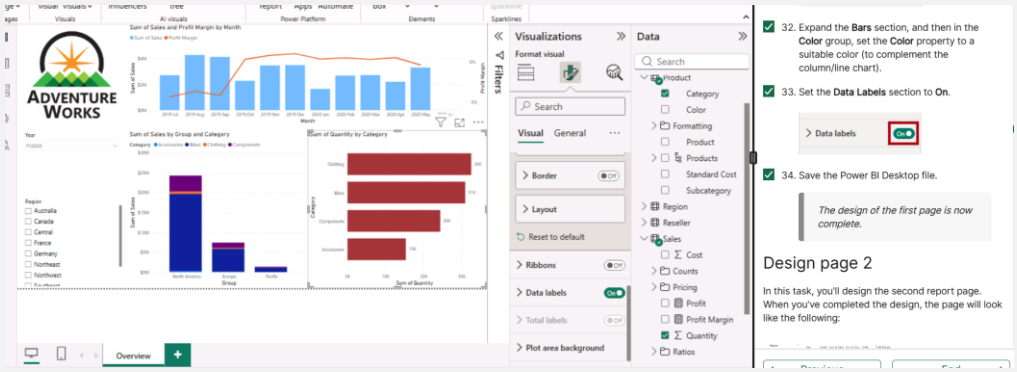
XP4180

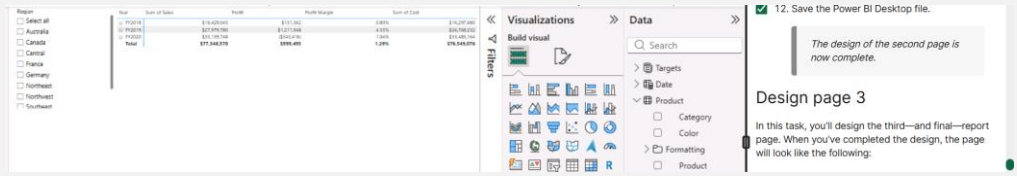
Day 4: Task 1

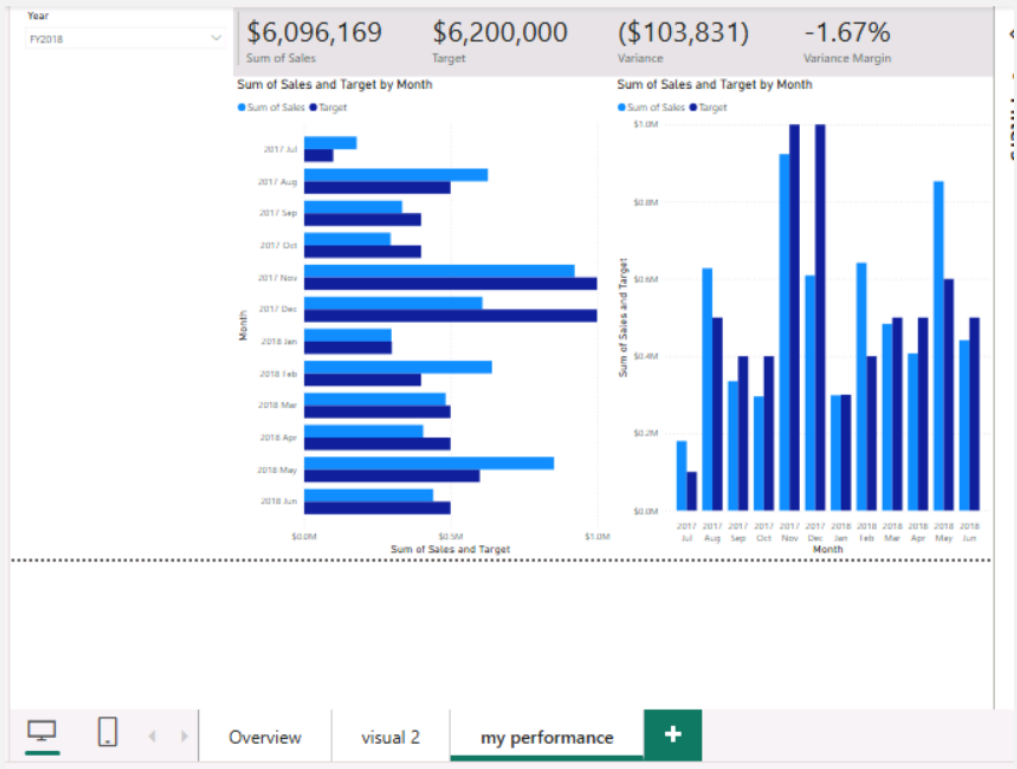
Please complete **Design Power BI reports** on Skillable. Once complete, paste a print screen below and in the collaboration board.

1) Visit https://leeptalent.learnondemand.net/
2) Sing into your profile
5) Go to **PL-300T00-A Design and Manage Analytics Solutions Using Power BI**
3) Launch and complete **Lab 8: Design Power BI reports.**

Paste your completed lab here







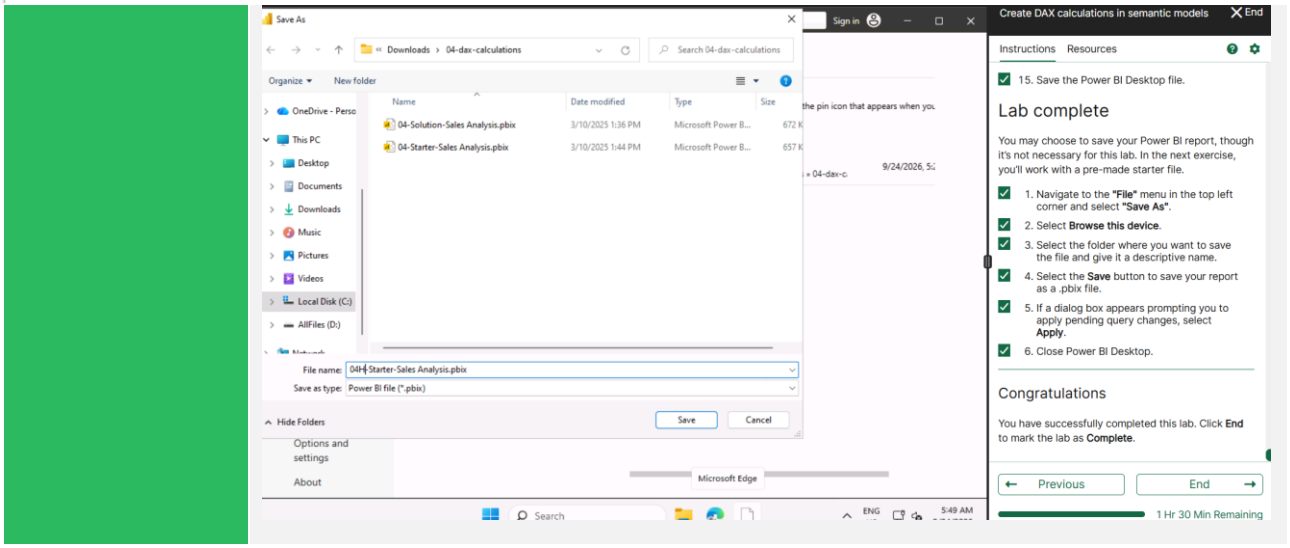
Day 4: Task 2

Please complete **Create DAX calculations in semantic models** on Skillable. Once complete, paste a print screen below and in the collaboration board.

1) Visit https://leeptalent.learnondemand.net/
2) Sing into your profile
6) Go to **PL-300T00-A Design and Manage Analytics Solutions Using Power BI**
3) Launch and complete **Lab 4: Create DAX calculations in semantic models.**

Paste your completed lab here

Salesperson	Sum of Sales	Target	Variance	Variance Margin
Amy Alberts	$10,288,626	$19,450,000	-$9,161,374	-47.10%
Brian Welcker	$77,548,570	$221,700,000	-$144,151,430	-65.02%
David Campbell	$12,004,822	$19,625,000	-$7,620,178	-38.83%
Garrett Vargas	$13,875,633	$23,675,000	-$9,799,367	-41.39%
Jae Pak	$8,410,883	$13,575,000	-$5,164,117	-38.04%
Jillian Carson	$7,633,387	$13,675,000	-$6,041,613	-44.18%
José Saraiva	$13,875,633	$18,875,000	-$4,999,367	-26.49%
Linda Mitchell	$25,634,503	$40,850,000	-$15,215,497	-37.25%
Lynn Tsoflias	$1,391,025	$3,210,000	-$1,818,975	-56.67%
Michael Blythe	$21,987,348	$31,150,000	-$9,162,652	-29.41%
Pamela Ansman-Wolfe	$30,005,939	$53,850,000	-$23,844,061	-44.28%
Rachel Valdez	$1,877,743	$4,125,000	-$2,247,257	-54.48%
Ranjit Varkey Chudukatil	$4,527,840	$9,050,000	-$4,522,160	-49.97%
Shu Ito	$18,001,116	$59,850,000	-$41,848,884	-69.92%
Stephen Jiang	$65,868,919	$110,150,000	-$44,281,081	-40.20%
Syed Abbas	$1,391,025	$3,050,000	-$1,658,975	-54.39%
Tete Mensa-Annan	$12,004,822	$17,100,000	-$5,095,178	-29.80%
Tsvi Reiter	$7,638,607	$13,250,000	-$5,611,393	-42.35%
Total	**$77,548,570**			

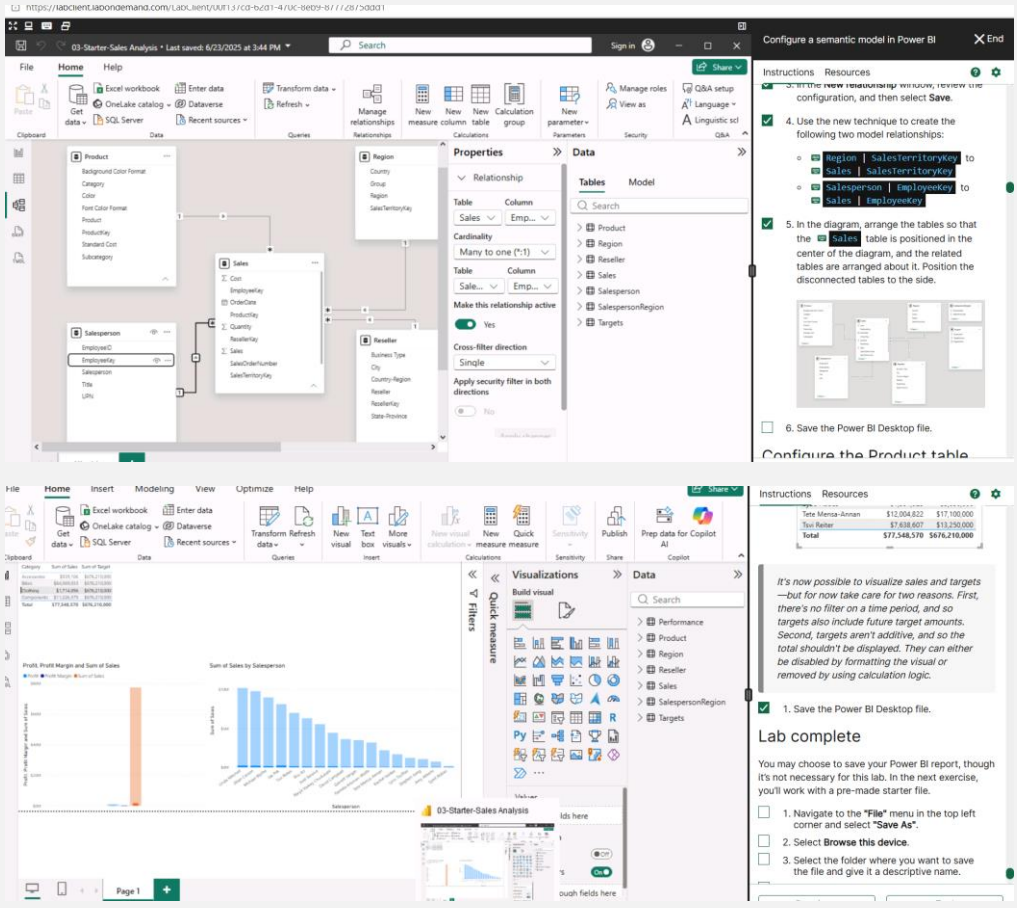

Save As
Downloads › 04-dax-calculations
Search 04-dax-calculations
Organize
New folder
Name
Date modified
Type
Size
04-Solution-Sales Analysis.pbix
3/10/2025 1:36 PM
Microsoft Power B...
672 K
04-Starter-Sales Analysis.pbix
3/10/2025 1:44 PM
Microsoft Power B...
657 K
OneDrive - Perso
This PC
Desktop
Documents
Downloads
Music
Pictures
Videos
Local Disk (C:)
AllFiles (D:)
File name: 04-Starter-Sales Analysis.pbix
Save as type: Power BI file (*.pbix)
Hide Folders
Save
Cancel
Options and settings
About
Microsoft Edge
Search
ENG
5:49 AM
Sign in
the pin icon that appears when you
9/24/2026, 5:
» 04-dax-c
Create DAX calculations in semantic models
End
Instructions
Resources
15. Save the Power BI Desktop file.
Lab complete
You may choose to save your Power BI report, though it's not necessary for this lab. In the next exercise, you'll work with a pre-made starter file.
1. Navigate to the "File" menu in the top left corner and select "Save As".
2. Select Browse this device.
3. Select the folder where you want to save the file and give it a descriptive name.
4. Select the Save button to save your report as a .pbix file.
5. If a dialog box appears prompting you to apply pending query changes, select Apply.
6. Close Power BI Desktop.
Congratulations
You have successfully completed this lab. Click End to mark the lab as Complete.
Previous
End
1 Hr 30 Min Remaining

Day 4: Task 3

Please complete **Configure a semantic model in Power BI.** on Skillable. Once complete, paste a print screen below and in the collaboration board.

1) Visit https://leeptalent.learnondemand.net/
2) Sing into your profile
3) Go to Lab **PL-300T00-A Design and Manage Analytics Solutions Using Power BI**
4) Launch and complete **Lab 3: Configure a semantic model in Power BI.**

Paste your completed lab here



Course Notes

It is recommended to take notes from the course, use the space below to do so, or use the revision guide shared with the class:

Additional Resources to learn and Practice

1) Full Beginner Project in Tableau | Tableau Tutorials for Beginners

2) Learn Tableau Basic Calculations (Calculated Fields) in Tableau (with 10+ examples) | Part 1
3) Tableau Hands-on Challenges Trailmix | Salesforce Trailhead
4) Introduction to Tableau - DataCamp Learn
5) Creating Dashboards in Tableau - DataCamp Learn
6) Sample Data | Tableau Public
7) How-To Videos | Tableau Public
8) Analyzing Data in Tableau - DataCamp Learn
9) Creating Dashboards in Tableau - DataCamp Learn
10) Introduction to Power BI - DataCamp Learn
11) Power BI Cheat Sheet | DataCamp
12) Data Modeling in Power BI - DataCamp Learn
13) Introduction to DAX in Power BI - DataCamp Learn
14) Data Visualization in Power BI - DataCamp Learn
15) **PL 300 Course on MS Learn.** Course PL-300T00-A: Design and manage analytics solutions using Power BI - Training | Microsoft Learn
This is an intermediate course.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer.